Precision and Recall

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One coherent metric-and-priority record.

Trace precision,recall and accuracy,interpret absent denominators and defend a conditional choice.

Keep one habit: W1–W3/W5 → selected new matrix → project threshold record → fresh changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the **recall** before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

Accuracy and Imbalance

My remembered idea, example and correction

Scores and Calibration Intuition

My remembered idea, example and correction

Decision Trees

My remembered idea, example and correction

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main clear/blocked matrix[[12,4], [1,3]]:TP3,FP4,FN1,TN12,precision3/7,recall3/4,accuracy15/20. No predicted positives[[16,0], [4,0]]:precision unavailable,recall0/4. No reference positives[[16,4],[0,0]]:recall unavailable. Picture TP6,FP2,FN4,TN8 gives6/8precision and6/10recall;moving2FNtoTPgives8/10forbothwithdifferentquestionmeanings.

  Selected independent matrix[[8,2],[3,6]]has19records,precision6/8,recall6/9,accuracy14/19.

- Reference for this exercise: open this week's student lesson, "Precision and Recall", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked matrix, give predicted-positive and reference-positive totals.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main clear/blocked matrix[[12,4], [1,3]]:TP3,FP4,FN1,TN12,precision3/7,recall3/4,accuracy15/20. No predicted positives[[16,0], [4,0]]:precision unavailable,recall0/4. No reference positives[[16,4],[0,0]]:recall unavailable. Picture TP6,FP2,FN4,TN8 gives6/8precision and6/10recall;moving2FNtoTPgives8/10forbothwithdifferentquestionmeanings.

  Selected independent matrix[[8,2],[3,6]]has19records,precision6/8,recall6/9,accuracy14/19.

- Reference for this exercise: open this week's student lesson, "Precision and Recall", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Calculate **precision**, **recall** and accuracy for the worked matrix.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main clear/blocked matrix[[12,4], [1,3]]:TP3,FP4,FN1,TN12,precision3/7,recall3/4,accuracy15/20. No predicted positives[[16,0], [4,0]]:precision unavailable,recall0/4. No reference positives[[16,4],[0,0]]:recall unavailable. Picture TP6,FP2,FN4,TN8 gives6/8precision and6/10recall;moving2FNtoTPgives8/10forbothwithdifferentquestionmeanings.

  Selected independent matrix[[8,2],[3,6]]has19records,precision6/8,recall6/9,accuracy14/19.

- Explain **precision** and **recall** for [[16, 0],[4, 0]].

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main clear/blocked matrix[[12,4], [1,3]]:TP3,FP4,FN1,TN12,precision3/7,recall3/4,accuracy15/20. No predicted positives[[16,0], [4,0]]:precision unavailable,recall0/4. No reference positives[[16,4],[0,0]]:recall unavailable. Picture TP6,FP2,FN4,TN8 gives6/8precision and6/10recall;moving2FNtoTPgives8/10forbothwithdifferentquestionmeanings.

  Selected independent matrix[[8,2],[3,6]]has19records,precision6/8,recall6/9,accuracy14/19.

- A team reports **recall** 100 percent but hides **precision** and counts. What is **miss**ing?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use clear/blocked on both axes, reference rows and prediction columns, with blocked positive. This supplied matrix has19records; do not use the20-case worked or pictured counts. State predicted-positive and reference-positive totals before dividing. No row IDs, score list or alternative threshold is supplied, so do not invent a threshold comparison for this independent task.

- Use [[8, 2],[3, 6]]. Give TP, FP, FN, **precision**, **recall** and accuracy.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Set a Plant Classifier review threshold and explain which error type it prioritizes and who makes the final decision.
- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.

- I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Calculate both ratios for **Precision recall** and **denominator**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for **Precision recall** and **denominator**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- TP=9,FP=3,FN=6,TN=12. Compute **precision** and **recall**. What happens to **precision** when a system predicts no positives?

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Positive is spotted: TP=8, FP=2, FN=4.
- Compute **precision** and **recall**.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- With disease positive, TP=6, FP=2 and FN=4. Calculate **precision** and **recall**, showing both **denominator**s.
- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 17 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or **miss**ing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- TP=9,FP=3,FN=6,TN=12. Compute **precision** and **recall**. What happens to **precision** when a system predicts no positives?

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Matrix convention for this exercise: rows are reference classes [clear, blocked]; columns are predicted classes [clear, blocked]. Blocked is the positive class, so the layout is [[TN, FP], [FN, TP]].

- Use [[8, 2],[3, 6]]. Give TP, FP, FN, **precision**, **recall** and accuracy.

My reasoning and evidence

▼ Extra practice · Pictured case

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Put **precision** and **recall denominator**s on the picture instead of memorising names.
- Positive means spotted. TP=6, FP=2, FN=4, TN=8 in twenty fictional leaves.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Were the original **precision** and **recall denominator**s the same? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/17

Keep precision and recall denominators distinct

	predicted spotted	predicted healthy
actual spotted	6	4
actual healthy	2	8

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/17

Precision and Recall

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Predicted positives

Eight leaves were flagged: precision=6/(6+2)=75%.

↓

2. Actual positives

Ten leaves were spotted: recall=6/(6+4)=60%.

↓

3. Keep goals separate

Precision asks how reliable flags are; recall asks how many spotted leaves were caught.

CHANGE ONE THING

Correct two false negatives into true positives.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Put **precision** and **recall denominator**s on the picture instead of memorising names.
- Which labelled step matches this detail? **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught.
- 1. Predicted positives
- 2. Actual positives
- 3. Keep goals separate
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Keep goals separate **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught. Read the picture in order.

Precision asks how reliable flags are; **recall** asks how many spotted leaves were caught. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Correct two false negatives into true positives. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**. TP becomes8 and FN 2; FP remains2, so the two **denominator**s happen to equal10.

Repair and keep the record

Review Accuracy and Imbalance (ai-4/16). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why can the 90% baseline still be unsuitable for a detection task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Scores and Calibration Intuition (ai-4/14). Goal: Interpret calibration using groups of forecasts. Short practice: Does a calibrated 0.8 score mean every such leaf is spotted?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Decision Trees (ai-4/11). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Which **input** distinguishes the original and changed tree?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Precision and Recall: the case

AI ACADEMY / AI-4/17 · 01 · LOOK AT THE FACTS

Precision and Recall: the case

The situation

Put precision and recall denominators on the picture instead of memorising names.

Supplied evidence

Positive means spotted. TP=6, FP=2, FN=4, TN=8 in twenty fictional leaves.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain precision and recall using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

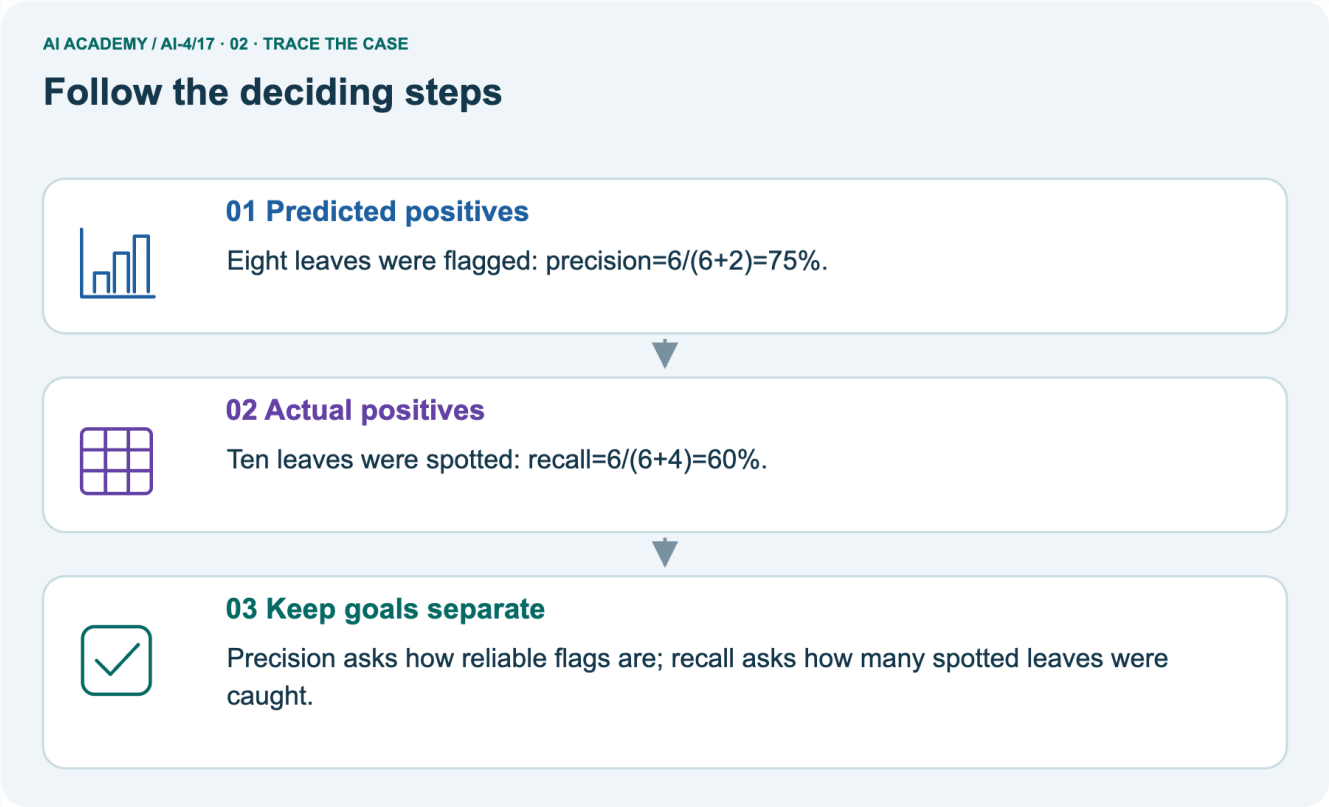
Put **precision** and **recall denominator**s on the picture instead of memorising names.
Supplied facts: Positive means spotted. TP=6, FP=2, FN=4, TN=8 in twenty fictional leaves.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



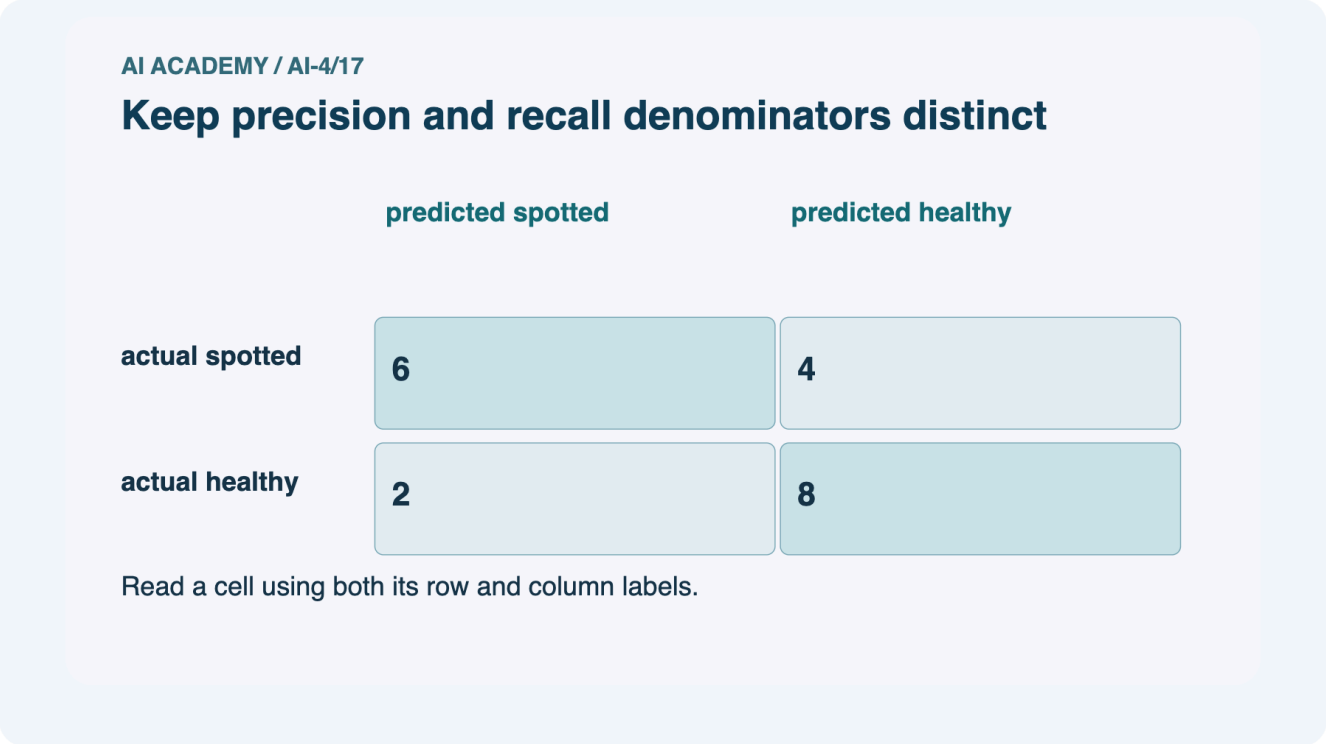
Follow the labelled picture with the matching explanation. Open the vector diagram

Predicted positives: Eight leaves were flagged: **precision**=6/(6+2)=75%. → Actual positives: Ten leaves were spotted: **recall**=6/(6+4)=**60%**. → Keep goals separate: **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Keep precision and recall denominators distinct



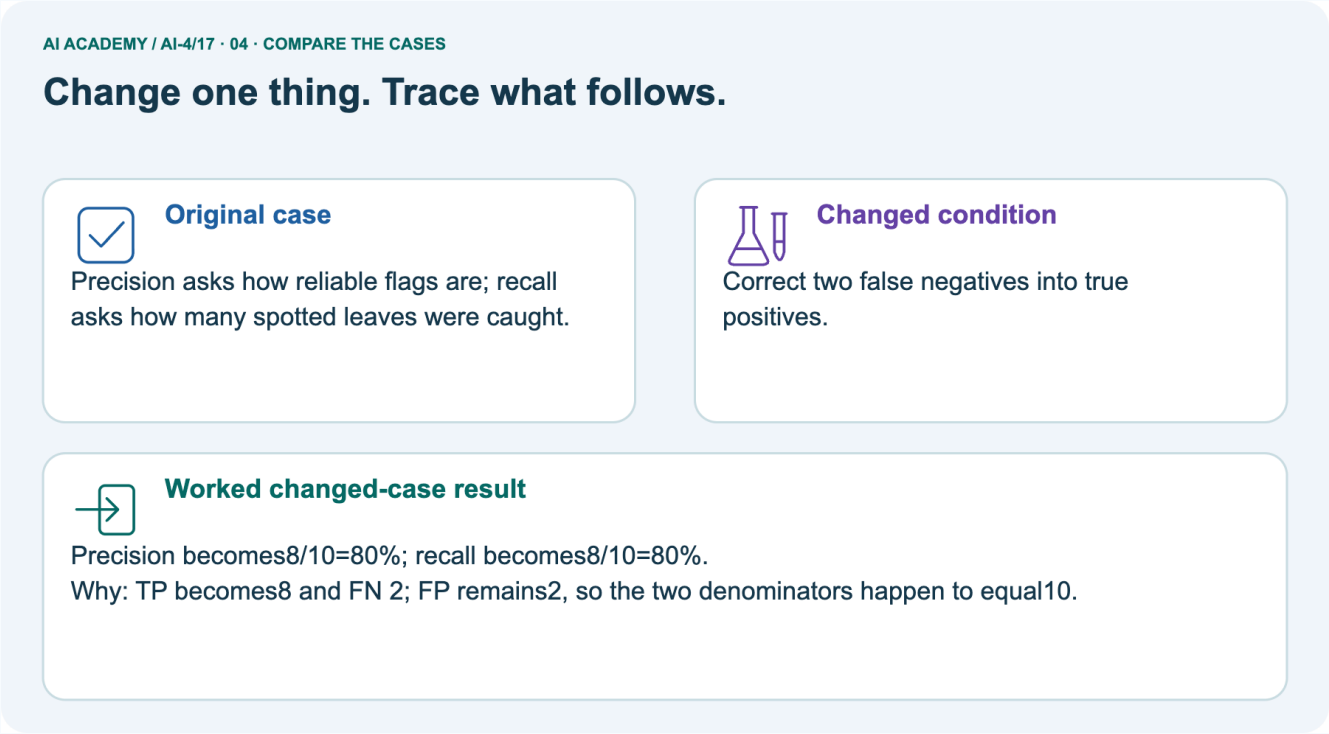
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep **precision** and **recall denominator**s distinct. actual spotted: predicted spotted = 6, predicted healthy = 4. actual healthy: predicted spotted = 2, predicted healthy = 8.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

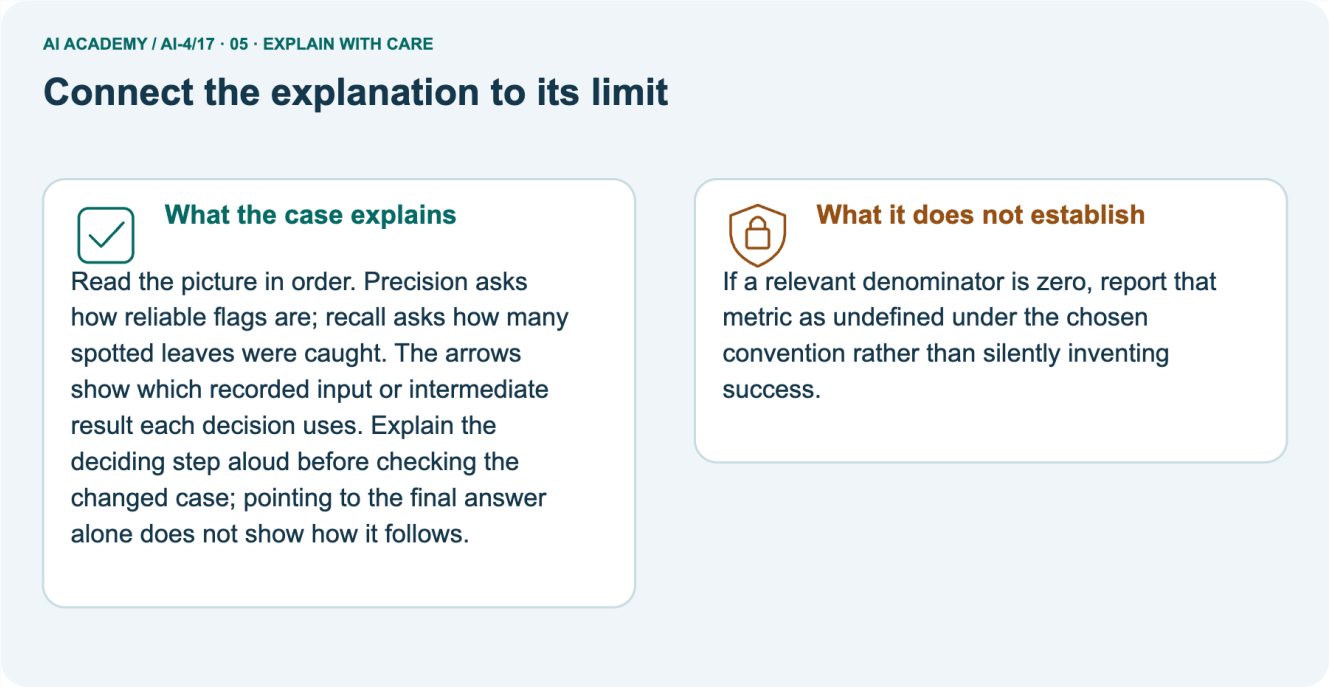
Original case: **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught. Changed condition: Correct two false negatives into true positives.

Expected result: **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**. Reason: TP becomes8 and FN 2; FP remains2, so the two **denominator**s happen to equal10.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught. The arrows show which recorded **input** or intermediate result each decision uses.

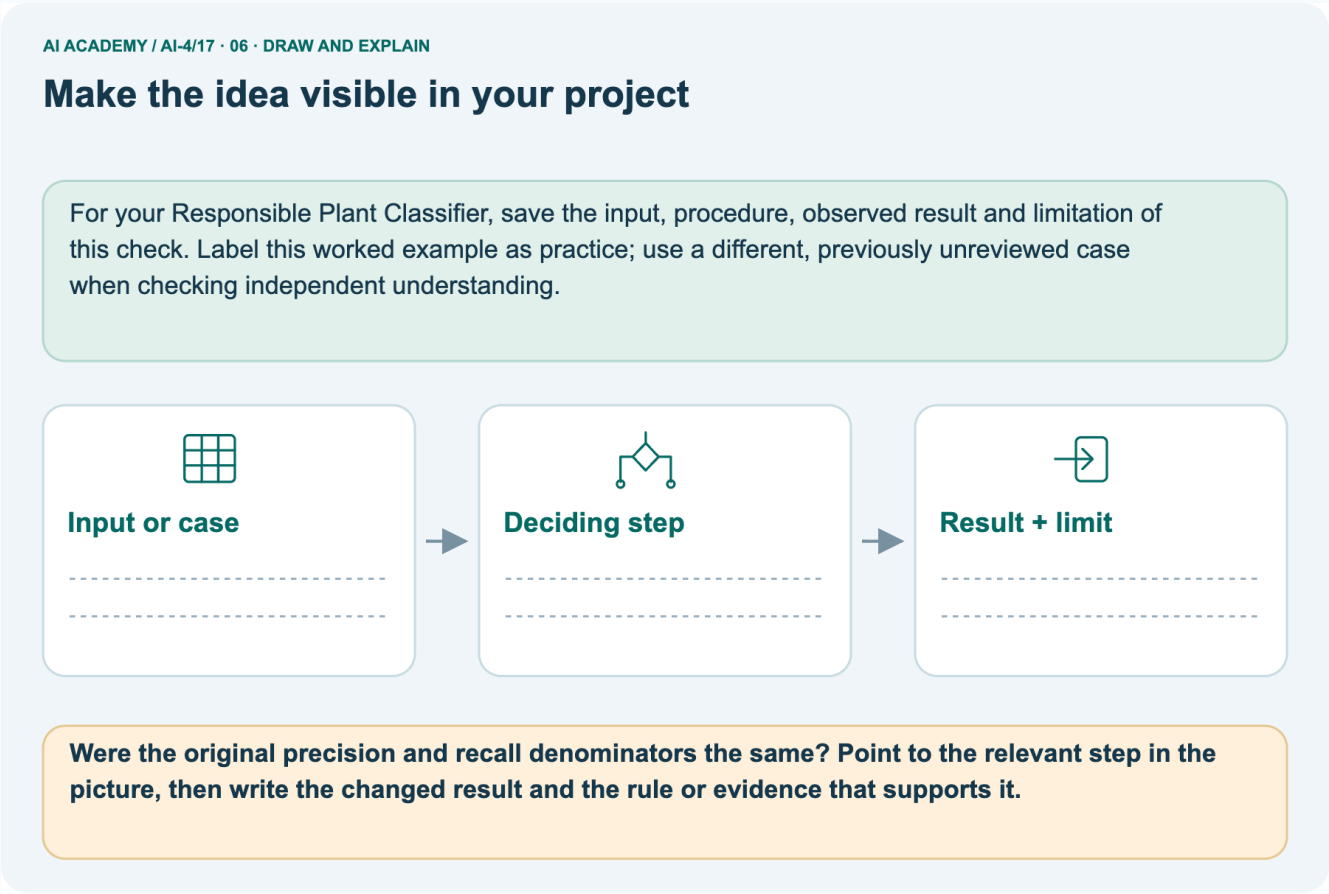
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: If a relevant **denominator** is zero, report that metric as **undefined** under the chosen convention rather than silently inventing success.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Were the original **precision** and **recall denominator**s the same? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Precision and Recall: the case

02. Follow the deciding steps

03. Keep precision and recall denominators distinct

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.